Area Between Curves

ANSWER KEY

Section 1: Area Between Curves Integrated with Respect to x

1 What is the area of the region enclosed by $y = x^2$ and $y = 4$?

a. $\frac{32}{3}$ ✓

b. $\frac{16}{3}$

c. 16

d. 8

Intersection points: $x^2 = 4 \Rightarrow x = \pm 2$. Area $= \int_{-2}^{2} (4 - x^2)dx = 2[4x - \frac{x^3}{3}]_0^2 = 2(8 - \frac{8}{3}) = 2(\frac{16}{3}) = \frac{32}{3}$.

2 Find the area between $y = x$ and $y = x^2$.

a. $\frac{1}{6}$ ✓

b. $\frac{1}{3}$

c. $\frac{1}{2}$

d. $\frac{1}{12}$

Intersections: $x = x^2 \Rightarrow x = 0, 1$. Over $[0, 1]$, $x \geq x^2$. Area $= \int_0^1 (x - x^2)dx = [\frac{x^2}{2} - \frac{x^3}{3}]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$.

3 Which integral represents the area enclosed by $y = \sin x$ and $y = \cos x$ from $x = 0$ to $x = \frac{\pi}{4}$?

a. $\int_0^{\pi/4} (\cos x - \sin x)dx$ ✓

b. $\int_0^{\pi/4} (\sin x - \cos x)dx$

c. $\int_0^{\pi/4} (\cos x + \sin x)dx$

d. $\int_0^{\pi/2} \cos x dx$

On $[0, \pi/4]$, $\cos x \geq \sin x$. The upper curve is $\cos x$ and lower is $\sin x$, so $\int_0^{\pi/4} (\cos x - \sin x)dx$.

Section 2: Area Between Curves Integrated with Respect to y

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4 Find the area of the region bounded by $x = y^2$ and $x = y + 2$.

a. $\frac{9}{2}$ ✓

b. $\frac{16}{3}$

c. $\frac{7}{2}$

d. 6

Intersections in y: $y^2 = y + 2 \Rightarrow y^2 - y - 2 = 0 \Rightarrow (y-2)(y+1) = 0 \Rightarrow y = -1, 2$. Area $= \int_{-1}^{2} (y + 2 - y^2)\,dy = [\frac{y^2}{2} + 2y - \frac{y^3}{3}]_{-1}^{2} = (2 + 4 - \frac{8}{3}) - (\frac{1}{2} - 2 + \frac{1}{3}) = \frac{10}{3} - (-\frac{7}{6}) = \frac{20+7}{6} = \frac{27}{6} = \frac{9}{2}$.

5 When is it advantageous to integrate with respect to y rather than x?

a. **When the boundary curves are expressed as $x = f(y)$ and solving for y requires multiple piecewise branches** ✓

b. When the area is negative

c. Always for quadratic curves

d. Never on the AP Exam

Integrating with respect to y (right curve minus left curve) simplifies the setup whenever solving for y yields plus/minus roots and splits the region into multiple integrals.

6 Find the area bounded by $x = 0, y = 1, y = 3,$ and $x = \frac{1}{y}$.

a. $\ln 3$ ✓

b. $\ln 2$

c. 3

d. 2

Area $= \int_{1}^{3} \frac{1}{y}\,dy = [\ln|y|]_{1}^{3} = \ln 3 - \ln 1 = \ln 3$.

Section 3: Intersecting Curves with Multiple Regions

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7 The graphs of $y = x^3$ and $y = x$ enclose how many symmetric regions?

a. 2 **symmetric regions ✓**

b. 1 region

c. 3 regions

d. Infinitely many

Intersections: $x^3 - x = 0 \Rightarrow x(x-1)(x+1) = 0 \Rightarrow x = -1, 0, 1$. The curves enclose 2 symmetric regions (on $[-1, 0]$ and $[0, 1]$).

8 What is the total area enclosed by $y = x^3$ and $y = x$?

a. $\frac{1}{2}$ ✓

b. $\frac{1}{4}$

c. 0

d. 1

Total Area $= 2\int_0^1 (x - x^3)dx = 2[\frac{x^2}{2} - \frac{x^4}{4}]_0^1 = 2(\frac{1}{2} - \frac{1}{4}) = 2(\frac{1}{4}) = \frac{1}{2}$.

9 Why does $\int_{-1}^{1} (x - x^3)dx = 0$ NOT represent the enclosed area?

a. **The integrand is odd, causing positive and negative signed areas to cancel out ✓**

b. The curves do not intersect at 0

c. The integral diverges

d. The bounds are incorrect

A definite integral computes net signed area. For true area, one must integrate $|f(x) - g(x)|$.

Section 4: Area Partitions & Horizontal/Vertical Lines

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10 The line $x = k$ divides the region between $y = x^2$ and the x-axis from $x = 0$ to $x = 2$ into two equal areas. Find k.

a. $\sqrt[3]{4}$ ✓

b. 1

c. $\sqrt{2}$

d. $\frac{4}{3}$

Total Area $= \int_0^2 x^2 dx = \frac{8}{3}$. Half area is $\frac{4}{3}$. $\int_0^k x^2 dx = \frac{k^3}{3} = \frac{4}{3} \Rightarrow k^3 = 4 \Rightarrow k = \sqrt[3]{4}$.

11 Find the area enclosed by $y = e^x, y = e^{-x},$ and the vertical line $x = 1$.

a. $e + \frac{1}{e} - 2$ ✓

b. $e - 1$

c. $e + \frac{1}{e}$

d. $2e - 2$

Intersections: $e^x = e^{-x} \Rightarrow x = 0$. On $[0, 1]$, $e^x > e^{-x}$. Area $= \int_0^1 (e^x - e^{-x})dx = [e^x + e^{-x}]_0^1 = (e + e^{-1}) - (1 + 1) = e + \frac{1}{e} - 2$.

12 Find the area of the region bounded by $y = \sqrt{x}$, the line $y = 2$, and the y-axis.

a. $\frac{8}{3}$ ✓

b. 4

c. $\frac{4}{3}$

d. 2

Integrating with respect to y: $x = y^2$ from $y = 0$ to $y = 2$. Area $= \int_0^2 y^2 dy = [\frac{y^3}{3}]_0^2 = \frac{8}{3}$.

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